

Multiple Choice (10 marks)

1. Which expression is equivalent to 4×9 ?
 - (a) $(4 \times 4) + (4 \times 5)$
 - (b) $(4+4) \times (4+5)$
 - (c) $(4+4)+(4+5)$
 - (d) $(4 \times 4) \times (4 \times 5)$
2. Suppose X and Y are random variables with $E(X) = 312$, $\text{var}(X) = 6$, $E(Y) = 307$, and $\text{var}(Y) = 8$. What are the expected value and variance of the random variable $X + Y$?
 - (a) $E(X + Y) = 619$, $\text{var}(X + Y) = 7$
 - (b) $E(X + Y) = 619$, $\text{var}(X + Y) = 10$
 - (c) $E(X + Y) = 619$, $\text{var}(X + Y) = 14$
 - (d) There is insufficient information to answer this question.
3. A coach rounded the number of runners at a track meet to the nearest 10. The rounded number of runners is 400. Which number could be the actual number of runners at the track meet?
 - (a) 382
 - (b) 397
 - (c) 406
 - (d) 447
4. What is 60% of 30?
 - (a) 1.8
 - (b) 18
 - (c) 180
 - (d) 1800
5. All of Bella's photos are in 8 photo books. Each photo book has 25 photos. Which expression shows how to find the total number of photos Bella has?
 - (a) $25 \div 8$
 - (b) $25 - 8$
 - (c) $25 + 8$
 - (d) 25×8

True / False (10 marks)

Answer True or False

6. True or False: The solution to the equation $x + 71 = -22$ is $x = 49$.
7. True or False: The solution to the equation $13/4 = x/7$ is $x = 22.75$.
8. True or False: The central limit theorem states that the sample mean is normally distributed only if the sample size is small and the population is normally distributed.
9. True or False: The sum of 110 and 70 is 180.
10. True or False: A worker can produce 220 parts in 35 hours if she takes 7 hours to produce 22 parts.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Find the greatest value of b such that $-b^2 + 7b - 10 \geq 0$.
12. Consider a sequence of three real numbers that forms an arithmetic progression with a first term of 9. If 2 is added to the second term and 20 to the third term, these numbers then form a geometric progression. Analyze the conditions of this scenario and determine the smallest possible value for the third term of the resulting geometric progression. Provide a detailed explanation of your reasoning and calculations.

End of Paper